

LAB 2 • DATA PREPARATION

Transform the data and build the model

Clean by meaning, not appearance. Preserve real negatives and percentages, remove only invalid records, and connect the tables with a predictable star schema.

35–40 minutes Power Query + Model view Deliverable: validated model

Transformation order

Remove invalid rows

Trim + standardize

Apply locale types

Validate counts

Relate tables

Order matters: remove rows that do not belong before spending time converting and correcting their values.

1. Budget vs Actuals

Source text

"(\$120,000)" * 6.00% * N/A

Typed result

-120000 * 0.06 * null

Promote the source headers

1. If the columns are named Column1, Column2, and so on, select **Home** → **Use first row as headers**.
2. Confirm the first headers are Program ID, Program Name, Fiscal Year, and Quarter.
3. Apply the same step to Program and Purchase Orders if their headers were not promoted during import.

The screenshot shows the Power Query Editor window. The 'Home' tab is selected, displaying various data transformation options. The 'Queries' pane on the left lists three queries: 'Budget vs Actuals', 'Program', and 'Purchase Orders'. The main area shows a data table with the following columns and rows:

	Program ID	Program Name	Branch	Contract Number
1	PRG-001	Thunderbolt Sustainment Support	Air Force	FA8101-24-C-00...
2	PRG-002	Cyber Defense Initiative	Army	W91CRB-24-C-0...
3	PRG-003	Satellite Communications Upgrade	Space Force	FA8808-24-C-00...
4	PRG-004	Logistics Command Platform	Navy	N00024-24-C-55...
5	PRG-005	Next-Gen Radar Systems	Air Force	FA8730-24-C-00...

Promote headers before cleaning

Use the first CSV row as field names before filtering, selecting, or typing columns.

Keep data rows and remove the duplicate

1. Filter **Program ID** with **Text Filters** → **Begins With** → **PRG-**. This safely removes blank and notes rows even if the footer grows.
2. Select all columns and choose **Home** → **Remove Rows** → **Remove Duplicates**.
3. Confirm the query now contains **120 rows**.

Table.TransformColumnTypes(#"Promoted headers", {"Program ID", type text}, {"Program Name", type text})

	Program ID	Program Name	Fiscal Year	Quarter	Cost Center	Budget Category
1	PRG-001				CC-4100	Labor
2	PRG-001				CC-4100	materials
3	PRG-001				CC-4100	Travel
4	PRG-001				CC-4100	Sub-Contracts
5	PRG-001					
6	PRG-001					
7	PRG-001					
8	PRG-001					
9	PRG-002					
10	PRG-002					
11	PRG-002					
12	PRG-002					
13	PRG-002					
14	PRG-002					
15	PRG-002					
16	PRG-002					
17	PRG-003					
18	PRG-003					
19	PRG-003					
20	PRG-003					

Sort ascending
Sort descending
Remove empty
Text filters

Search

- (Select all)
- (blank)
- Notes: Data exported from D365 on 04/15/20...
- PRG-001
- PRG-002
- PRG-003
- PRG-004
- PRG-005

OK Cancel

Equals...
Does not equal...
Begins with...
Does not begin with...
Ends with...
Does not end with...
Contains...
Does not contain...
In...
Not in...

Keep valid program records

Use Text Filters → Begins With and enter PRG- rather than deleting a fixed number of footer rows.

Table.SelectRows(#"Changed column type", each Text.StartsWith([Program ID], "PRG"))

	Program ID	Program Name	Fiscal Year	Quarter	Cost Center	Budget Category	Budgeted Amount	Actual Amount	Variance	Variance %	Funding Source
1	PRG-001	Thunderbolt Sustainment Support	FY2025	Q1	CC-4100	Labor	2,500,000.00	\$2,350,000	\$150,000	6.00%	DoD Appropriation
2	PRG-001	Thunderbolt Sustainment Support	FY2025	Q1	CC-4100	materials	1,800,000.00	\$1,920,000	(\$120,000)	-6.70%	DoD Appropriation
3	PRG-001	Thunderbolt Sustainment Support	FY2025	Q1	CC-4100	Travel	350,000.00	\$280,000	\$70,000	20.00%	DoD Appropriation
4	PRG-001	Thunderbolt Sustainment Support	FY2025	Q1	CC-4100	Sub-Contracts	4,200,000.00	\$4,150,000	\$50,000	1.20%	DoD Appropriation
5	PRG-001	Thunderbolt Sustainment Support	FY2025	Q2	CC-4100	Labor	2,500,000.00	\$3,900,000	(\$1,400,000)	-56.00%	DoD Appropriation
6	PRG-001	Thunderbolt Sustainment Support	FY2025	Q2	CC-4100	Materials	1,800,000.00	\$1,750,000	\$50,000	2.80%	DoD Appropriation
7	PRG-001	Thunderbolt Sustainment Support	FY2025	Q2	CC-4100	travel	350,000.00	\$580,000	(\$230,000)	-65.70%	DoD Appropriation
8	PRG-001	Thunderbolt Sustainment Support	FY2025	Q2	CC-4100	Subcontracts	4,200,000.00	\$5,800,000	(\$1,600,000)	-38.10%	DoD Appropriation
9	PRG-002	Cyber Defense Initiative	FY2025	Q1	CC-5200	Labor	1,200,000.00	\$750,000	\$450,000	37.50%	DARPA Grant
10	PRG-002	Cyber Defense Initiative	FY2025	Q1	CC-5200	Materials	400,000.00	\$380,000	\$20,000	5.00%	DARPA grant
11	PRG-002	Cyber Defense Initiative	FY2025	Q1	CC-5200	Travel	150,000.00	\$145,000	\$5,000	3.30%	DARPA Grant
12	PRG-002	Cyber Defense Initiative	FY2025	Q1	CC-5200	Sub-Contracts	800,000.00	\$790,000	\$10,000	1.30%	DARPA Grant
13	PRG-002	Cyber Defense Initiative	FY2025	Q2	CC-5200	Labor	1,200,000.00	\$780,000	\$420,000	35.00%	DARPA Grant
14	PRG-002	Cyber Defense Initiative	FY2025	Q2	CC-5200	Materials	400,000.00	\$420,000	(\$20,000)	-5.00%	DARPA Grant
15	PRG-002	Cyber Defense Initiative	FY2025	Q2	CC-5200	Travel	150,000.00	\$130,000	\$20,000	13.30%	DARPA Grant
16	PRG-002	Cyber Defense Initiative	FY2025	Q2	CC-5200	Subcontracts	800,000.00	\$810,000	(\$10,000)	-1.30%	DARPA Grant
17	PRG-003	Satellite Comm Upgrade	FY2025	Q1	CC-3300	Labor	3,100,000.00	\$3,050,000	\$50,000	1.60%	Space Force
18	PRG-003	Satellite Comm Upgrade	FY2025	Q1	CC-3300	Materials	2,200,000.00	\$2,400,000	(\$200,000)	-9.10%	Space Force
19	PRG-003	Satellite Comm Upgrade	FY2025	Q1	CC-3300	Travel	200,000.00	\$180,000	\$20,000	10.00%	Space Force
20	PRG-003	Satellite Communications Upgrade	FY2025	Q1	CC-3300	Sub-Contracts	5,500,000.00	\$7,400,000	(\$1,900,000)	-34.50%	Space Force

Copy preview data
Remove columns
Remove other columns
Add column from examples...
Remove duplicates
Remove errors
Replace values...
Replace errors...
Merge columns
Change type
Transform columns
Group by...
Fill
Unpivot columns
Unpivot only selected columns

Remove the repeated record

Select all columns before choosing Remove duplicates so only an exact duplicate is removed.

Trim and normalize text

1. Select Program ID, Program Name, Cost Center, Budget Category, Funding Source, Contract Number, and Status.

2. Choose **Transform** → **Format** → **Trim**.
3. Capitalize Each Word for Budget Category and Status.
4. In Budget Category, replace Sub-Contracts with Subcontracts.

The screenshot shows the Power Query Editor interface. The 'Transform' tab is active, and the 'Format' menu is open, displaying options like 'lowercase', 'UPPERCASE', 'Capitalize Each Word', 'Trim', and 'Clean'. The 'Trim' option is highlighted. In the background, a table is visible with columns 'Program ID' and 'Program Name'. The 'Queries' pane on the left shows 'Budget vs Actuals', 'Program', and 'Purchase Orders'.

Program ID	Program Name	Fiscal Year	Quarter
1	PRG-001	Thunderbolt Sust	Q1
2	PRG-001	Thunderbolt Sust	Q1
3	PRG-001	Thunderbolt Sust	Q1
4	PRG-001	Thunderbolt Sust	Q1
5	PRG-001	Thunderbolt Sust	Q2
6	PRG-001	Thunderbolt Sust	Q2
7	PRG-001	Thunderbolt Sust	Q2
8	PRG-001	Thunderbolt Sust	Q2
9	PRG-001	Thunderbolt Sust	Q2
10	PRG-001	Thunderbolt Sust	Q2

Trim selected text columns

Ctrl-select the text columns, then use Transform → Format → Trim to remove leading and trailing spaces in one step.

The screenshot shows the Power Query Editor interface. The 'Transform' tab is active, and the 'Replace values' option is selected. The 'Queries' pane on the left shows 'Budget vs Actuals'. The background table is the same as in the previous screenshot.

Program ID	Program Name	Fiscal Year	Quarter
1	PRG-001	Thunderbolt Sust	Q1
2	PRG-001	Thunderbolt Sust	Q1
3	PRG-001	Thunderbolt Sust	Q1
4	PRG-001	Thunderbolt Sust	Q1
5	PRG-001	Thunderbolt Sust	Q2
6	PRG-001	Thunderbolt Sust	Q2
7	PRG-001	Thunderbolt Sust	Q2
8	PRG-001	Thunderbolt Sust	Q2
9	PRG-001	Thunderbolt Sust	Q2
10	PRG-001	Thunderbolt Sust	Q2

Open Replace values

Select Budget Category, then choose Replace values.

Replace values

Replace one value with another in the selected columns.

☒ Basic ☐ Advanced

Value to find

Sub-Contracts

Replace with

Subcontracts

OK

Cancel

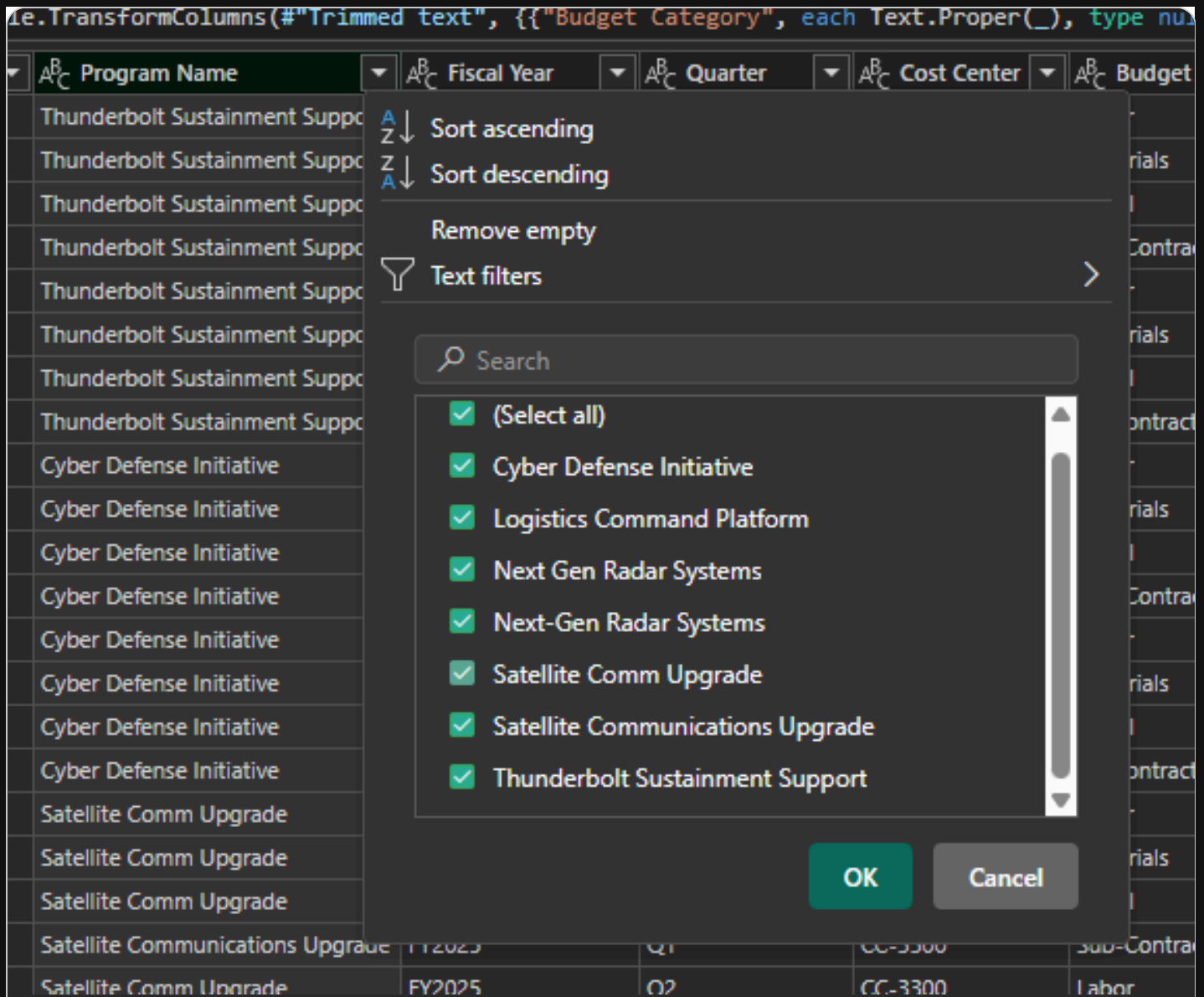
Standardize the category

Replace Sub-Contracts with Subcontracts and repeat for any remaining category variants.

Number column		Date and time column				
ory", each Text.Proper(_), type nullable text}}, {"Status", each Text.Proper(_), type nullable text}})						
Quarter	ABC Cost Center	ABC Budget Category	\$ Budgeted Amount	ABC Actual Amount	ABC Variance	ABC
	CC-4100	Labor				6.00
	CC-4100	Materials				-6.70
	CC-4100	Travel				20.00
	CC-4100	Sub-Contracts				1.20
	CC-4100	Labor				-56.00
	CC-4100	Materials				2.80
	CC-4100	Travel				-65.70
	CC-4100	Subcontracts				-38.70
	CC-5200	Labor				37.50
	CC-5200	Materials				5.00
	CC-5200	Travel				3.30
	CC-5200	Sub-Contracts				1.30
	CC-5200	Labor				35.00
	CC-5200	Materials				-5.00
	CC-5200	Travel				13.30
	CC-5200	Subcontracts				-1.30
	CC-3300	Labor				1.60
	CC-3300	Materials				-9.10
	CC-3300	Travel				10.00
	CC-3300	Sub-Contracts				-34.50
	CC-3300	Labor				-3.20
	CC-3300	Materials				4.50

Inspect category values

Open the filter list to confirm which labels still represent the same category.



Inspect program-name variants

The source contains shortened and standardized names; the next step creates one governed value for each program.

Standardize program names with a conditional column

1. Select **Add column** → **Conditional column**.
2. Name the new column **ProgramName**.
3. Add the rule: if Program Name equals Next Gen Radar Systems, output Next-Gen Radar Systems.
4. Add an Else if rule: if Program Name equals Satellite Comm Upgrade, output Satellite Communications Upgrade.
5. For Else, choose the **Program Name** column so every already-valid name passes through unchanged.
6. Select OK and verify that ProgramName contains only the five governed names.

Power Query Editor

Home

Transform

Add column

View

Help



Column from examples



Custom column



Invoke custom function



Conditional column



Index column



Rank column



Duplicate column



Format



Measures



ABC



abc

Queries [3]



Budget vs Actuals



Program



Purchase Orders

Open Conditional column

Use Add column → Conditional column to encode both name corrections in one auditable step.

Add conditional column

Add a conditional column that is computed from the other columns or values.

New column name *

ProgramName

Column name *

Program Name

Operator *

equals

Value

Next Gen Radar Systems

Then

Next-Gen Radar Syste...

Else if

Program Name

equals

Satellite Comm Upgrade

Then

Satellite Communicati...

Add clause

Else

Program Name

OK

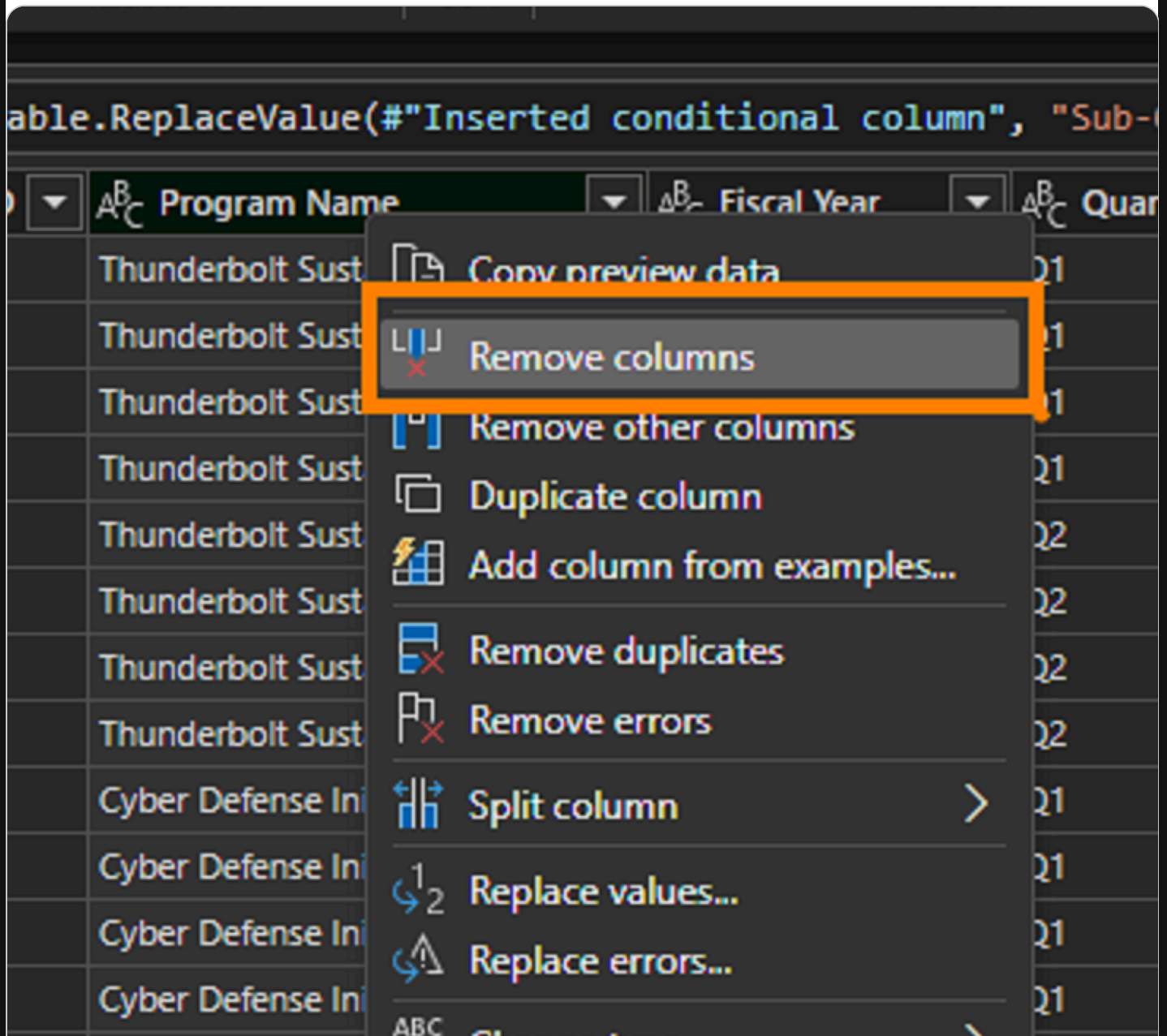
Cancel

Define the correction rules

Return the standardized name for each known variant and use the original Program Name column as the Else output.

Replace the original name column

1. Right-click the original **Program Name** column and choose **Remove columns**.
2. Rename **ProgramName** to **Program Name**.
3. Set its data type to **Text**.
4. Open the filter list and confirm exactly five program names remain.



Remove the old column

Delete the inconsistent source column only after the conditional result has been verified.

ABC	Status	Program Name
	On Track	Thunderbolt Sustainment Support
	Over Budget	Thunderbolt Sustainment Support
	On Track	Thunderbolt Sustainment Support
	On Track	Thunderbolt Sustainment Support
	Over Budget	Thunderbolt Sustainment Support
	On Track	Thunderbolt Sustainment Support
	Over Budget	Thunderbolt Sustainment Support
	Over Budget	Thunderbolt Sustainment Support
	On Track	Cyber Defense Initiative
	On Track	Cyber Defense Initiative
	On Track	Cyber Defense Initiative
	On Track	Cyber Defense Initiative
Restore the governed field name Rename ProgramName to Program Name and keep its type as Text.		

```

: Communications Upgrade" else [Program Name])

```

Contract Number

Status

ProgramName

A Z ↓ Sort ascending
Z A ↓ Sort descending

Remove empty

Text filters

>

Search

☒ (Select all)

☒ Cyber Defense Initiative

☒ Logistics Command Platform

☒ Next-Gen Radar Systems

☒ Satellite Communications Upgrade

☒ Thunderbolt Sustainment Support

OK

Cancel

Verify the result

Convert money without losing negative values

1. In Actual Amount, Variance, and Variance %, replace N/A with an empty value.
2. Select Budgeted Amount, Actual Amount, and Variance.
3. Choose **Transform** → **Data Type** → **Using Locale**.
4. Use **Currency** and **English (United States)**. Parentheses must remain negative: (\$120,000) becomes -120000.

Convert the percentage correctly

1. Select Variance % and choose **Data Type** → **Using Locale**.
2. Use **Percentage** and **English (United States)**.
3. Verify 6.00% becomes 0.06, displayed as 6% after formatting.

Correction from the old guide: do not remove the percent sign and then convert to Decimal. That produces 6, which Power BI formats as 600%. Also do not strip parentheses without first preserving the negative sign.

Add a reporting period

1. Select **Add Column** → **Custom Column**.
2. Name it **Fiscal Period** and use [Fiscal Year] & " " & [Quarter].
3. This creates sortable labels such as FY2025 Q1 and FY2026 Q2 for the trend chart.

2. Purchase Orders

Deduplicate and trim

1. Select PO ID, then **Home** → **Remove Rows** → **Remove Duplicates**. Expect **20 rows**.
2. Select every text column and apply **Transform** → **Format** → **Trim**.

Clean amount and date types

1. In Amount Paid, replace NULL and - with 0.
2. Select PO Amount, Amount Paid, and Amount Remaining; use **Data Type** → **Using Locale** → **Currency** → **English (United States)**.
3. Select Order Date and Delivery Date; use **Data Type** → **Using Locale** → **Date** → **English (United States)**.
4. Check Column Quality. Do not replace date errors with null until you inspect the original value and confirm it is invalid.

Normalize business labels

1. Capitalize Each Word in PO Status and Priority.
2. Replace Apex Dynamics Corp. with Apex Dynamics Corp and Centurion Inc. with Centurion Inc.
3. Replace Satellite Comm Upgrade with Satellite Communications Upgrade.

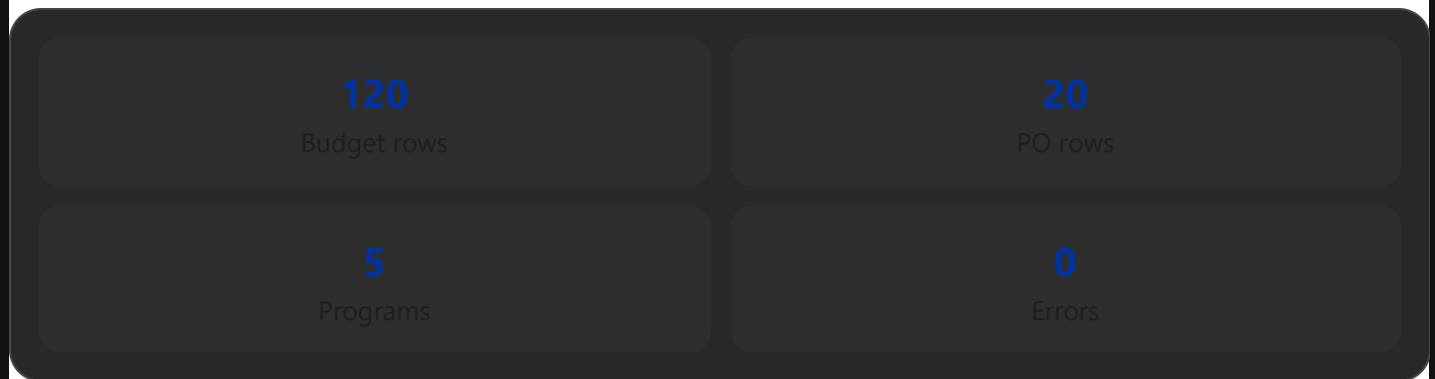
3. Program

Protect the dimension key

1. Trim all four columns.
2. Set Program ID, Program Name, Branch, and Contract Number to **Text**.
3. Confirm Program ID is unique: 5 rows and 5 distinct values.

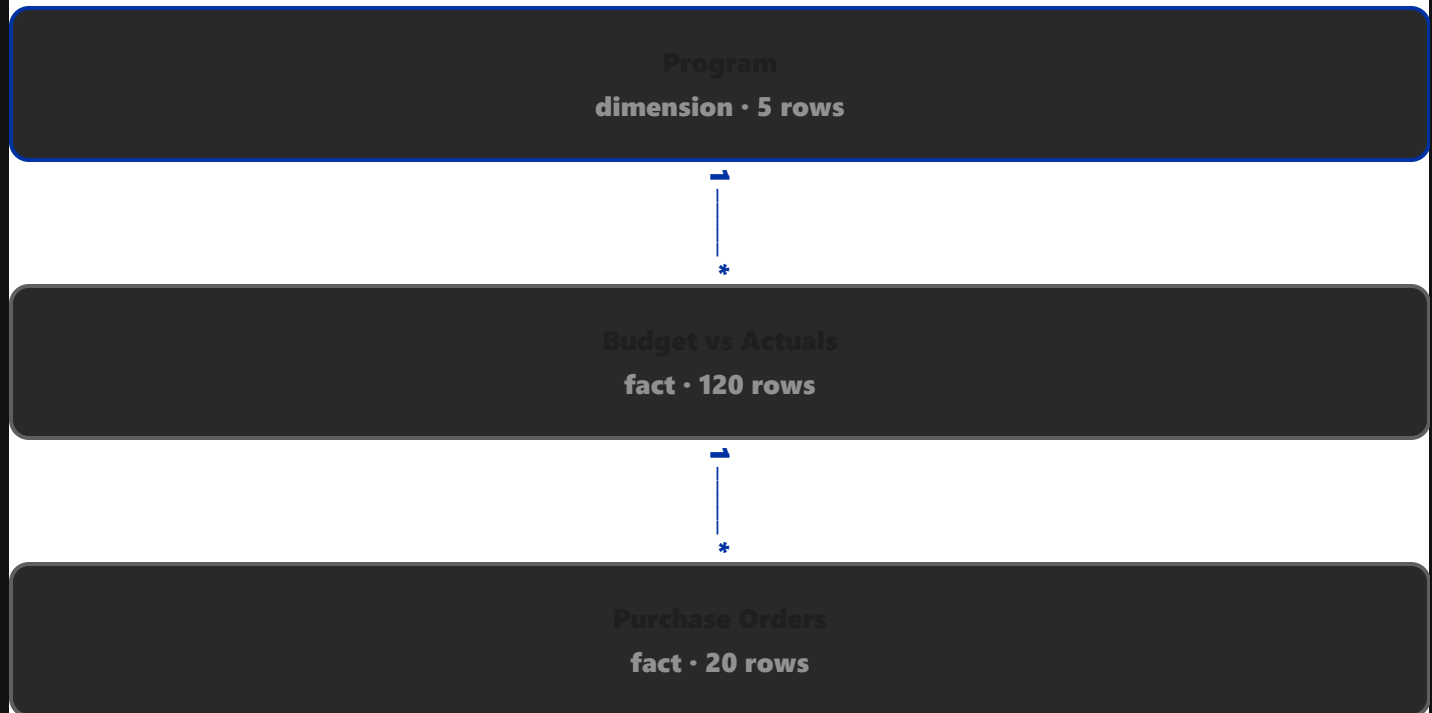
Modeling rule: use **Program[Program Name]** in slicers and axes. The fact-table names were intentionally inconsistent; the dimension is the governed label.

Validate before loading



1. Set Column Profiling to the entire dataset.
2. Confirm amount columns show numeric distributions, not text values.
3. Confirm Variance % is Percentage and dates are Date.
4. Select **Home** → **Close & Apply**.

Build the star schema



1. Open **Model view**.
2. Relate **Program[Program ID]** to **Budget vs Actuals[Program ID]**.
3. Relate **Program[Program ID]** to **Purchase Orders[Program ID]**.
4. For both relationships use **One-to-many (1:*)**, **Single** cross-filter direction, and make the relationship active.
5. Hide Program Name in both fact tables to steer report authors toward the Program dimension.

Single-direction filters flow from Program to both facts. This lets one Program slicer filter budget and procurement without ambiguous loops.